

Modelling Day, Night and the Seasons

Javalab Day and Night, Seasons, White Night, Polar Night · Javalab

Years 7-10

~30 minutes

Science · Earth & Space Science

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://javalab.org/en/daylight_en/

Curriculum links: AC9S7U03 · AC9S10U04

Learning intentions

- I can explain how Earth's rotation causes day and night.
- I can use a model of Earth's tilt and orbit to explain why seasons occur.
- I can describe unusual daylight phenomena, such as white nights and polar nights, using the model.

Getting started

Open the simulation and set the location to your own approximate latitude first. Note today's day length before you start changing any settings.

Tasks

1. Record the day length (hours of daylight) shown for your chosen latitude on today's date.

2. Change the date to the summer solstice, then the winter solstice, keeping the latitude the same. Record the day length for each.

3. Using what the model shows about Earth's tilt, explain why day length changes between the solstices at your chosen latitude.

4. Keeping the date fixed, change the latitude across three values (e.g. near the equator, a mid-latitude, near a pole) and record the day length each time.

Latitude	Day length

5. Based on your table, what happens to the difference between summer and winter day length as latitude increases toward the poles?

6. Move the latitude slider until you find a location experiencing a 'white night' or 'polar night'. Explain, using Earth's tilt and rotation, why this phenomenon occurs there and not at your home latitude.

7. In one or two sentences, explain why day and night are caused by Earth's rotation, while seasons are caused by axial tilt combined with orbit — two separate effects, not the same cause.

Extension (optional)

Use the model to predict what day length patterns would look like at every latitude if Earth had no axial tilt at all. Explain your reasoning.
